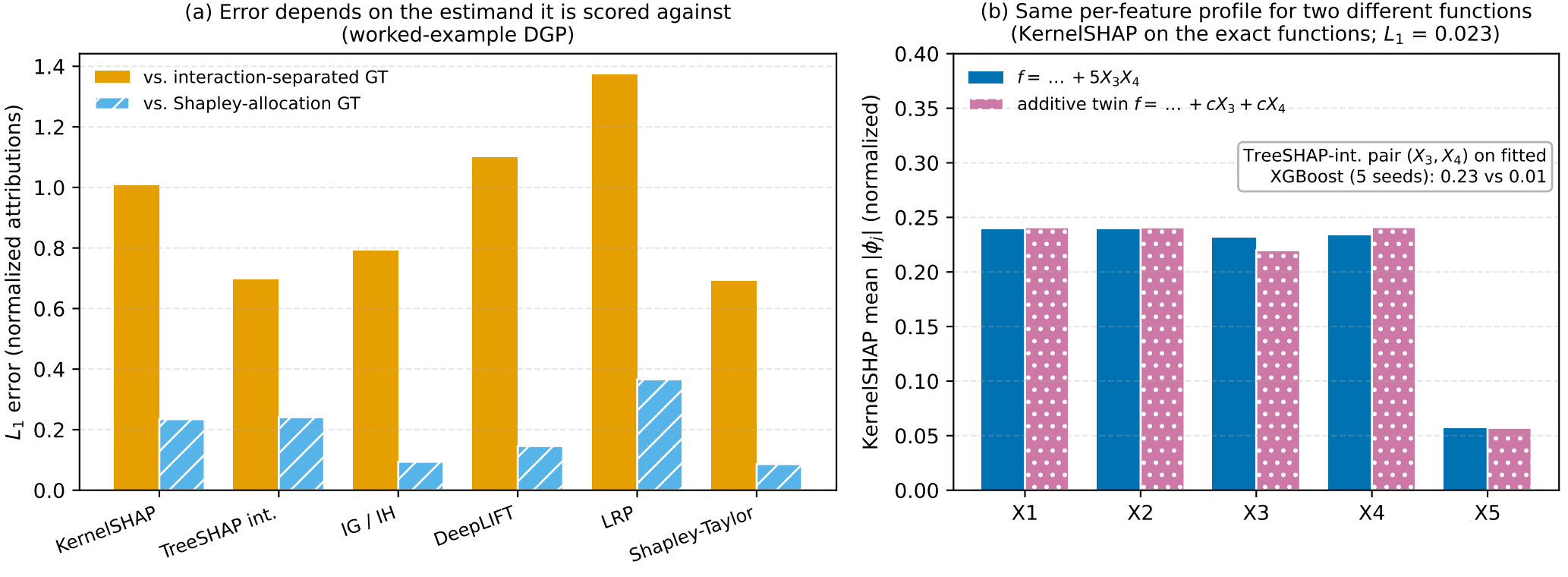
(a) Error depends on the estimand it is scored against
(worked-example DGP)
vs. interaction-separated GT
vs. Shapley-allocation GT
L_1 error (normalized attributions)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
KernelSHAP
TreeSHAP int.
IG / IH
DeepLIFT
LRP
Shapley-Taylor
(b) Same per-feature profile for two different functions
(KernelSHAP on the exact functions; L_1 = 0.023)
$f = \ldots + 5X_3X_4$
additive twin $f = \ldots + cX_3 + cX_4$
TreeSHAP-int. pair (X_3, X_4) on fitted
XGBoost (5 seeds): 0.23 vs 0.01
KernelSHAP mean $|\phi_j|$ (normalized)
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
X1
X2
X3
X4
X5